

introduction

- definition of emergence -> higher level properties coming into existence when multiple separate components are scaled, eg. ant swarm
- more is different -> individual behaviour might be unpredictable/same but when a lot of them are scaled different global behaviour emerges, eg all ants travel in random directions but overall they are able to move the object through the maze
- less is more -> when we know that an emergent phenomenon exists, easier to use a higher level theory that ignores microscopic details rather than use a lower dimension theory that has millions of variables/data points, eg. each ants behaviour is harder to track but as a colony it can be studied
- LLM lit and scaling laws support certain abilities arising suddenly when neurons are scaled to deep nns -> shows emergent behavior
- questions: what is emergence, why it matters -> covered above, it matters as it leads to cost saving by predicting easier higher level theory of generalisation and without it we would have to observe the separate microscopic interactions and components rendering the process tedious.
- evidence for emergence -> new representations being formed, abstractions leading to better efficiency
- what is emergent though?? capabilities vs intelligence!! --> **MAIN POINT: LLMs have emergent capabilities but not intelligence**
- eg. calculator -> it can solve the engineered tasks that it was made for but cannot intelligently form new capabilities.

primer on emergence

- 1. same higher level theory is guaranteed for emergence to be there, coarse graining or ignorance of less important features is key, eg given is molecular vs fluid dynamics (local -> less effective vs global predictions-> more effective)
- emergence claim = coarse graining (efficient selection) + compression of system (comes at free cost as a result to energy minimization via less is more.

conditions for emergence:

1. scaling: new capabilities become suddenly visible or a new internal organisation gets formed due to either scaling data or compute
2. criticality: rapid changes are observed in the behaviour, the new formed organisation changes its description to something else

3. compression: the models internal representation is compressed to increase efficiency
4. novel bases: new methods for better representation learning are found.
5. generalisation: capabilities of solving a completely new task B emerging when training is done on task A

types of emergence : knowledge in vs knowledge out

- knowledge out emergence: if a complex behaviour emerges from the local interactions of multiple individual components for free. simple rule following at scale = completely different behaviour is seen. eg. colony of ants moving food through an obstruction
- knowledge in emergence: instead of simple rule following individual agents if the separate entity is complex and are trained/evolved from the information present in the environment. when these interact together a different type of emergence phenomenon is observed. eg LLMs -> neurons interacting at different layers

framework to analyse emergence

emergence through scaling

eg1. double descent - it occurs in large llms where loss first goes down then overfits and then again goes down is said to be emergent in nature through scaling but even linear and polynomial regression show the same phenomenon

hence not emergent

eg2. ON THE UNIVERSALITY OF NEURAL ENCODINGS IN CNNs paper - they show that the peak of the double descent appears synonymously with the change in the internal representations of the model where it starts to hold only important features and discards the other ones -> finds a more novel base for representation learning.

therefore this is emergent as it shows -> scaling induces phase transition taking place due to novel bases being discovered

- main argument that is presented with scaling and emergence -> as you scale the data and params there are discontinuous jumps observed in accuracy of certain tasks much like phase transition compared with physical transitions of water, however they say this comparison is unfair as we don't whether the jump is actually discontinuous in nature or just appears so due to selection of that specific portion or task
- with loss functions we are implicitly programming a certain property through scaling training therefore it cannot be termed as emergent inherently.

emergence through compression

eg1. othello gpt -> where the model creates a compressed internal world model and emergence is claimed to refer to the creation of a coarse-grained internal representations. The problem with this was however the internal wm formed was more like a "bag of heuristics" rather than having a coherent latent space representation

therefore cannot be termed emergent

emergence through novel bases and manifolds

manifolds serve as an efficient data representation method and emergence in this context would be having a newer elegant structures for these, however our limited understanding of how LLMs actually build and utilise these structures for their efficiency leads us to ambiguity and therefore any claims about this need more information for proper verification.

generalisation through emergence

emergence in its form can automatically give rise to generalisation over varied tasks through the formation of coarse grained higher level theory, again though there exist multiple frameworks to test the gaps of generalisation(VC dimension check, Probably Approximately Correct framework etc.) but we don't have enough task specific internal mechanisms and data to judge this fairly upon.

conclusions

- a lot of claims are called emergence on the basis of the misinterpreted meaning and with the number of variables that are included when we talk about the current LLMs are too many (eg.parameters, non linear interactions between components, size of training data etc.) hence for any specific claims the ability to actually place the finger on is limited and therefore cannot be verified. Using the word hence requires a higher level of understanding and must be done with more responsibility.

emergence claims judgement summarised

- eg. 3 digit addition-> not possible on 8b, 12b but gets 80% acc on 175b
- the current emergence definition in context of LLMs -> if they acquire sudden jump in capabilities/accuracy through scaling data/compute/params etc.
- **this is not the correct definition of emergent and therefore we need a scientific grounding of the word to avoid misinterpretation**
- double descent - it occurs in large llms where loss first goes down then overfits and then again goes down is said to be emergent in nature through scaling but even linear and polynomial regression show the same phenomenon
- othello gpt -> where the model creates a compressed internal world model and emergence is claimed to refer to the creation of a coarse-grained internal representations. The problem

with this was however the internal wm formed was more like a "bag of heuristics" rather than having a coherent latent space representation, therefore not emergent

- ON THE UNIVERSALITY OF NEURAL ENCODINGS IN CNNS paper - they show that the peak of the double descent appears synonymously with the change in the internal representations of the model where it starts to hold only important features and discards the other ones -> finds a more novel base for representation learning.
therefore this is emergent as it shows -> scaling induces phase transition taking place due to novel bases being discovered

to do

a nice experiment on the basis of this paper would be to give out an agent the constraints from the paper for quantifying emergence then give it a /goal for searching through all the relevant literature upto '26, a thorough output would depend on it checking through the criterias of coarse-grained internal representations, compression, novel bases/manifolds, phase-transition-like reorganization, causal use of the representation, and generalization in a check pass manner.

- the output: a list of the found works that qualify under the said constraints of emergence in the paper and why (in a brief note format)
- the input: codex with /goal + detailed AGENTS.md which forces the agent to check the abs + results first and then the mechanisms for evidence grounding and proper claim testing.
- ambitious goal: under turn of events maybe an automated pipeline where every new relevant paper gets passed through this sieve and added to a public community index which can serve as reference for the community and researchers.